

MBBS 214: Human Physiology II - Lecture Notes

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1. The Cardiac Cycle

The cardiac cycle is the sequence of electrical and mechanical events from the start of one heartbeat to the start of the next. At a heart rate of 75 beats per minute one cycle lasts 0.8 seconds, with ventricular systole lasting about 0.3 s and diastole about 0.5 s.

Phases of the cycle (left ventricle): (1) Atrial systole: the atria contract and add the final part of ventricular filling. (2) Isovolumetric contraction: ventricular pressure rises with all valves closed. The atrioventricular (AV) valves close, producing the first heart sound (S1). (3) Rapid ejection: when ventricular pressure exceeds aortic pressure (about 80 mmHg) the aortic valve opens and most of the stroke volume is ejected. (4) Reduced ejection: ejection slows as ventricular pressure falls. (5) Isovolumetric relaxation: the aortic valve closes, producing the second heart sound (S2) and the diastolic notch on the aortic pressure curve. Ventricular pressure falls with all valves closed. (6) Rapid filling: the AV valves open when ventricular pressure falls below atrial pressure. (7) Reduced filling (diastasis).

Key values: end-diastolic volume (EDV) about 120 mL; end-systolic volume (ESV) about 50 mL; stroke volume (SV) = EDV - ESV = 70 mL. Ejection fraction = SV / EDV = 70 / 120, about 58 percent (normal 55 to 70 percent). Cardiac output = heart rate x SV = 75 x 70 = 5250 mL/min, about 5 L/min. Aortic pressure is about 120/80 mmHg. Frank-Starling law: increased venous return raises EDV, which increases the force of contraction and therefore the stroke volume.

2. Renal Clearance and Glomerular Filtration

Glomerular filtration is driven by Starling forces. Net filtration pressure = glomerular capillary hydrostatic pressure (about 60 mmHg) - Bowman's capsule hydrostatic pressure (about 18 mmHg) - plasma oncotic pressure (about 32 mmHg) = about 10 mmHg. The glomerular filtration rate (GFR) is about 125 mL/min, or 180 L/day, in a healthy adult.

Renal clearance is the volume of plasma completely cleared of a substance per minute: $C = (U \times V) / P$, where U is the urine concentration, V is the urine flow rate and P is the plasma concentration. Inulin is freely filtered and neither reabsorbed nor secreted, so its clearance equals GFR. Worked example: P = 0.5 mg/mL, U = 25 mg/mL, V = 2.5 mL/min gives $C = (25 \times 2.5) / 0.5 = 125$ mL/min. Creatinine clearance approximates GFR in clinical practice.

Para-aminohippuric acid (PAH) is filtered and almost completely secreted, so its clearance estimates renal plasma flow (RPF), about 625 mL/min. Renal blood flow = RPF / (1 - haematocrit), about 1100 mL/min at a haematocrit of 0.45. Filtration fraction = GFR / RPF = 125 / 625 = 0.2 (20 percent). If the clearance of a substance is less than GFR there is net tubular reabsorption. If it is greater than GFR there is net tubular secretion.

Glucose is freely filtered and normally completely reabsorbed in the proximal tubule by sodium-glucose co-transporters (SGLT2 and SGLT1). The renal threshold for glucose is about 180 mg/dL (10 mmol/L). Above this level the transporters are saturated, the transport maximum (about 375 mg/min) is exceeded, and glucose appears in the urine (glycosuria), as in uncontrolled diabetes mellitus.

3. Action Potentials

The resting membrane potential is about -70 mV in neurons (about -90 mV in skeletal muscle). It is maintained by the Na⁺/K⁺ ATPase (3 Na⁺ out, 2 K⁺ in) and by the greater permeability of the resting membrane to K⁺. The equilibrium potential for K⁺ is about -90 mV and for Na⁺ about +60 mV.

Phases: (1) A stimulus depolarises the membrane to threshold, about -55 mV. (2) Rapid depolarisation: voltage-gated Na⁺ channels open and Na⁺ enters the cell in a positive feedback loop, reaching a peak of about +30 mV. (3) Repolarisation: Na⁺ channels inactivate and voltage-gated K⁺ channels open, so K⁺ leaves the cell. (4) After-hyperpolarisation: K⁺ channels close slowly, so the membrane is briefly more negative than rest. (5) Return to the resting potential.

All-or-none principle: once threshold is reached a full-size action potential fires. Stronger stimuli increase the frequency of firing, not the amplitude. Absolute refractory period: Na⁺ channels are inactivated and no second action potential can be triggered whatever the stimulus. Relative refractory period: some Na⁺ channels have recovered but K⁺ channels are still open, so only a stronger than normal stimulus can trigger an action potential.

Propagation: local current flow depolarises the next segment of membrane. In unmyelinated axons conduction is continuous and slow. In myelinated axons conduction is saltatory: the action potential jumps between nodes of Ranvier, where Na⁺ channels are concentrated, which is faster and uses less energy. Conduction velocity increases with axon diameter and myelination. Local anaesthetics work by blocking voltage-gated Na⁺ channels.

4. Respiratory Volumes and Capacities

Lung volumes are measured by spirometry. Typical adult values: tidal volume (TV) 500 mL; inspiratory reserve volume (IRV) 3000 mL; expiratory reserve volume (ERV) 1100 mL; residual volume (RV) 1200 mL. Capacities are sums of volumes: inspiratory capacity = TV + IRV = 3500 mL; functional residual capacity = ERV + RV = 2300 mL; vital capacity = IRV + TV + ERV = 4600 mL; total lung capacity = vital capacity + RV = 5800 mL. Spirometry cannot measure RV, FRC or TLC, which require helium dilution or body plethysmography.

Minute ventilation = TV x respiratory rate = 500 x 12 = 6000 mL/min. Anatomical dead space is about 150 mL. Alveolar ventilation = (TV - dead space) x respiratory rate = (500 - 150) x 12 = 4200 mL/min. Rapid shallow breathing lowers alveolar ventilation more than the minute ventilation suggests.

The FEV₁/FVC ratio is normally about 80 percent or higher. In obstructive disease (asthma, COPD) the FEV₁/FVC ratio is reduced, below 70 percent, and RV and TLC may rise. In restrictive disease (pulmonary fibrosis) the FVC and vital capacity are reduced while the FEV₁/FVC ratio is normal or increased.